Collective Intelligence 2nd Assignment

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Abstract

Important Note: I had to complete the assignment alone, because my two teammates informed me that they were either no longer part of the course or had other assignment, thus after consulting with Professor Zoltán Barta, as per his instructions I left out some parts of the assignment.

In this report, I describe the multi-agent experiments conducted using `LevelBasedForaging-v0`. I performed experiments with no communication and communication with vocab size 1 and 4. These were performed with both MAPPO and IPPO.

1 Overview and limitations

The notebook implements a two-agent reinforcement learning setup for `LevelBasedForaging-v0` from Jumanji, using PPO with generalized advantage estimation and compares the following:

- No-communication baseline: there is no communication between the agents
- Communication policy: the agents receive observations along with discrete messages
- Communication policy with reduced vocabulary size: the agents receive observations along with discrete messages, but the vocabulary size is reduced to 1 which is equivalent to constant messages

Limitations:

- The number of agents is hard-coded to be 2 for simplicity
- The messages have only been tested with vocabulary sizes 4 and 1
- As per my instructions, neither continuous communication protocol, nor an LSTM policy have been implemented

2 Environment

The base environment is created in the following way: `env = JumanjiEnv("LevelBasedForaging-v0", batch_size=[128])`

3 Policies

There are two policy variants. One without and one with communication.

3.1 No Communication Policy

This policy does not have any communication. The observations of the agents are mapped through a trunk and fed into a linear action head to produce the action logits. These are then sampled to create the actions.

3.2 Discreet Communication Policy

The communication is done with discreet messages that are made up of a vocabulary with size 4 (or 1). The agents output message and action logits which are fed into a torso with 2 heads, and action and a message head. The messages are being passed to the next step and an episode termination resets the messages to 0.

Messages are sampled from categorical distributions and the message entropy is also logged.

4 Critic

A critic network is constructed and evaluated with two different configurations, wither with MAPPO or IPPO.

Both versions are tested to identify the effect they have on learning.

5 Data Collection and Training

5.1 Data Collection

Training data is collected with `SyncDataCollector`. A `ReplayBuffer` stores the collected frames. Minibatches are sampled without replacement.

5.2 PPO Objective

The PPO loss is implemented via `ClipPPOLoss`, GAE is computed and appended to the data each iteration.

5.3 Training

Inside the training loop, the collected tensordict is flattened over the team dimension. The code then expands and reshapes `done`, `terminated`, and reward tensors to match the expected shapes for the PPO loss.

5.4 Optimization

Adam is used for optimization.

6 Experiments

Experiments were focused on determining whether communication was beneficial for the given task. In total, 6 different experiments were conducted, which were the following:

1. MAPPO critic (centralized):
    - No communication
    - Communication with vocabulary size 4
    - Communication with vocabulary size 1
2. IPPO critic (decentralized):
    - No communication
    - Communication with vocabulary size 4
    - Communication with vocabulary size 1

The code creates plots with the raw and rolling mean of returns per iteration. The window size used for the rolling mean is 10.

7 Results

The results of the experiment show that training is ineffective. There are no statistically significant separation of learning curves. I will elaborate below:

7.1 Learning Curves

The learning curves contain the mean episode reward plotted against the iterations counts. The plots are created for both the MAPPO and IPPO experiments. There are 6 lines on each plot. 3 of them are the no communication, communication with 4 vocabulary size and communication with 1 vocabulary size, and the 3 others are the rolling means of these respectively.

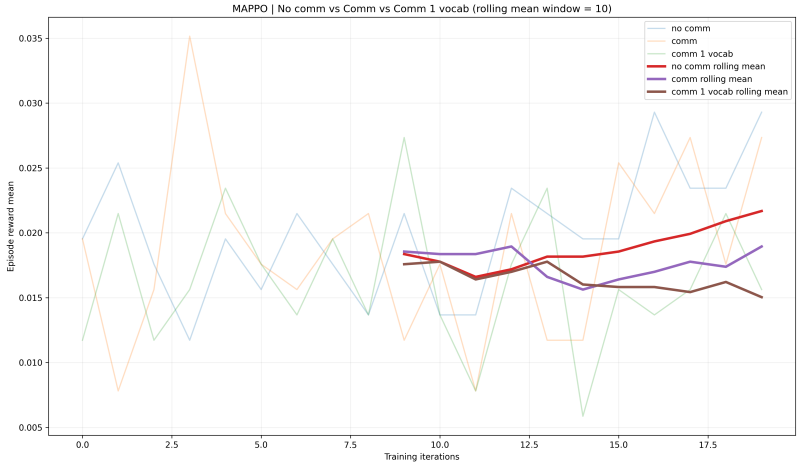


Figure 1: Learning curves of the MAPPO experiments

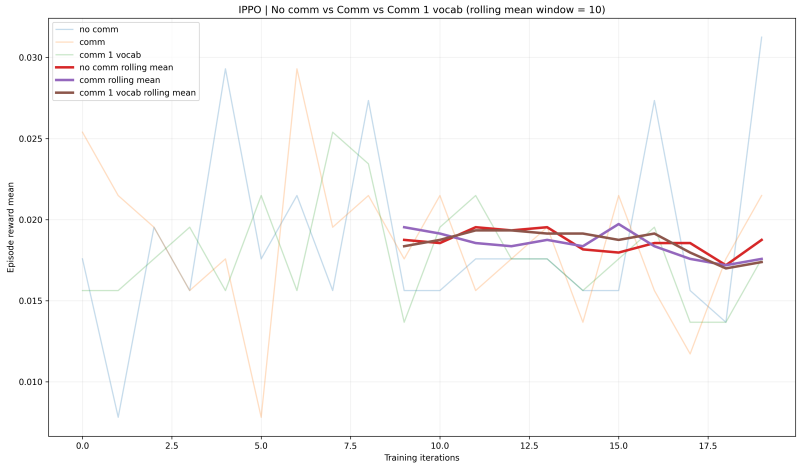


Figure 2: Learning curves of the IPPO experiments

In these plots it seems that there is no significant separation of the different configurations. The probable causes are explained below.

What should be seen is that since the no communication and the communication with vocabulary size 1 configurations are essentially the same, they should be together, while emergent

communication should be present with the communication with vocabulary size 4 configurations, meaning that mean episode rewards should steadily increase creating a distinct separation from the other two configurations.

7.2 Message Entropy

The below plots contain the entropies of the messages of both agents and their means. Because messages should only be meaningful with the communication with vocabulary size 4 configuration, only they are depicted.

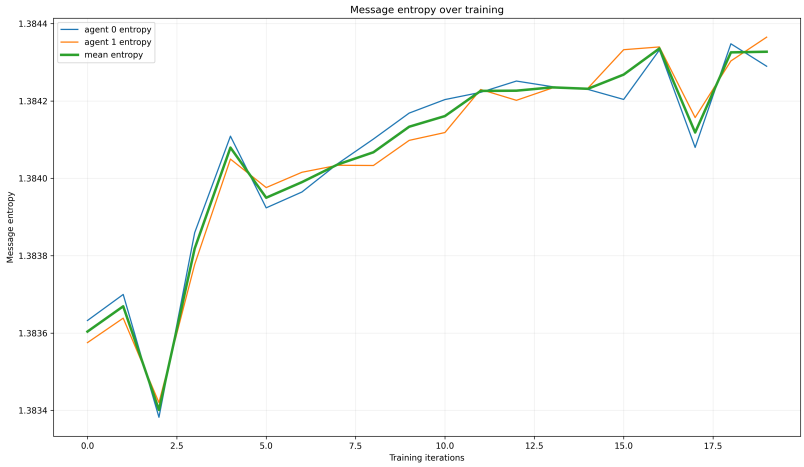


Figure 3: Message entropies of the MAPPO experiments

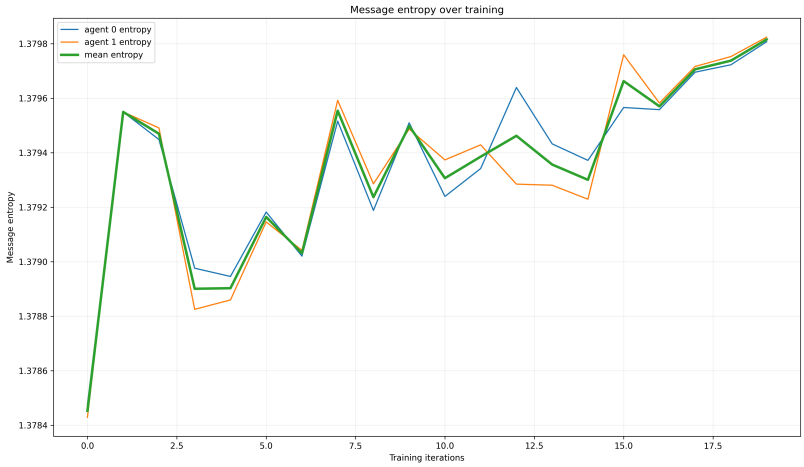


Figure 4: Message entropies of the IPPO experiments

What is expected is that the message entropy would decrease for a while, then stagnate and converge to some value. In contrast, we can see that here, the entropy is increasing, but in a relatively small quantity, meaning that the messages are basically noise, nothing is being learned.

7.3 Message probabilities

The plots here contain the message token usage probabilities. As with the entropies, only the communication with vocabulary size 4 configuration can be seen as only that experiment should have meaningful results, because of the size of the vocabulary. On the plots, there are 4 lines, corresponding to the 4 tokens of the vocabualry.

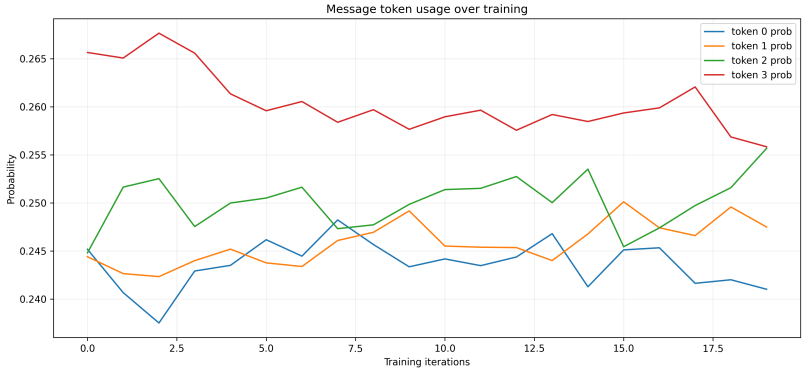


Figure 5: Message probabilities of the MAPPO experiments

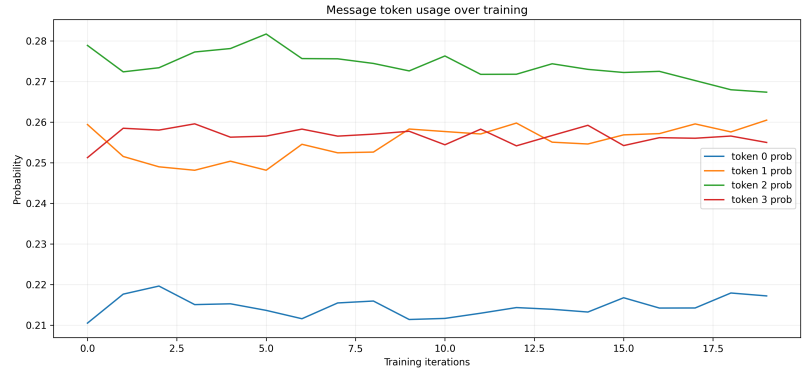


Figure 6: Message probabilities of the IPPO experiments

Again, the plots are inconclusive, or rather they show that token usage is basically random, with no improvements during training, meaning that the contents of the messages are most probably noise.

What should be seen here is that some tokens are being prioritized as training advances and 1, maybe 2 of them are being used almost exclusively.

7.4 Explanation of Results

The results of the experiments show that the agents have not improved at all with communication during and after training. This is possibly due to multiple factors:

- Faulty code: There may be some undetected fault in the code that causes learning to not function properly
- Learning is not incentivized, meaning that the the agents are not using communication, because it is not better than randomized actions

The most probable reason is that despite my best efforts, there is some fault in the code that makes the training essentially useless and thus the agents can never learn cooperation.

8 Reproducibility

Fixed seeds, and clearly separated hyperparameters ensure the reproducibility of the experiments.

I experimented with a lot of different hyperparameters looking for a combination of relatively low computation time while retaining as much separation and clarity in the results as possible.

The following hyperparameters were found to work the best:

Table 1: Key hyperparameters used

Parameter	Value
Batch size	128
Agents	2
Vocabulary size	4 (or 1)
Frames per batch	50,000
Iterations	20
Epochs per iteration	5
Minibatch size	4096
Learning rate	3×10^{-4}
PPO clip	0.2
gamma	0.99
lambda	0.95
Entropy eps.	0.02
Max steps per episode	500

9 Areas of Improvement

There are a multitude of ways the code, notebook and the experiments themselves could be improved, but were not implemented because of time-constraints:

- Implementing continuous embedding protocol
- Implementing LSTM policy
- Include more evaluation metrics
- Create a dockerfile and hydra configs for easier management and reproducibility
- Implement curriculum learning